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Kinetics of rouleaux formation using TV image analyzer. I. Human erythrocytes

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SHIGA, TAKESHI, KAZUHIKO IMAIZUMI, NOBORU HARADA, AND MISUZU SEKIYA. *Kinetics of rouleaux formation using TV image analyzer. I. Human erythrocytes.* Am. J. Physiol. 245 (Heart Circ. Physiol. 14): H252–H258, 1983.—An apparatus for determining the velocity of erythrocyte rouleaux formation was constructed, combining an inverted microscope, a transparent cone-plate viscometer, a TV image analyzer, and a computer. At lower shear rates, the overall process is the sedimentation and the rouleaux formation followed by the development of three-dimensional aggregates. The individual erythrocyte could be observed and the process was expressed by the time courses of the changes in the count and area of particles; taking the computed increment in the area/count, the rate of rouleaux formation could be estimated. The effects of shear rates, hematocrits, plasma proteins, and pH were quantified. The rate of rouleaux formation in autologous plasma increased by (*1*) lowering the shear rates ($1.9 \leq \dot{\gamma} \leq 15\ s^{-1}$), *2*) increasing the hematocrit (up to 0.6%), *3*) adding human fibrinogen (up to 600 mg/dl) or γ-globulin, and *4*) increasing pH. The transformation to echinocytes or to stomatocytes decreased the rate of rouleaux formation. The pH effect was explained by the increase in mean corpuscular volume at lower pH rather than by the changes in the electrostatic repulsion or in the protein binding.

aggregation; erythrocytes (human); fibrinogen; shear rate; pH dependence; image processing

THE DEVELOPMENT OF ERYTHROCYTE aggregation in the circulation is an important phenomenon with regard to its effect on blood viscosity at low shear and on sludge formation in the microcirculation (2, 6, 7, 12, 16, 19, 24). It is useful to be able to predict the risk of aggregation in vivo on the basis of in vitro experiments. However, the determinant factors for erythrocyte aggregation are numerous and complex, e.g., the applied shear stress, the plasma constituents, and the erythrocyte properties (3). Therefore, it is necessary to have knowledge of the basic kinetics of erythrocyte aggregation and of the effect of the numerous determinant factors.

To study kinetics of erythrocyte aggregation in vitro, the time courses of the growth of aggregates in plasma under certain shear rate regions must be recorded. For this purpose the individual erythrocytes must be observed separately and the number of erythrocytes in each aggregate counted. The "Rheoscope" developed by Schmid-Schönbein (23, 26), which consists of an inverted microscope and a transparent cone-plate visocometer, is an appropriate apparatus for such studies. Therefore, we combined a rheoscope with a TV image analyzer (TV camera and image processor) and a computer. The analyzer encoded the counts and areas of particles in microscopic TV images then transferred the data to a computer for data processing. The time courses of changes in the observed counts and areas of particles and in the computed area/count were plotted automatically.

The advantages of our apparatus are the automated data processing and the simplicity of data presentation, thus the present method is adequate for kinetic studies of the aggregation process. On the other hand, it is hard to study the whole blood because the hematocrit must be reduced considerably to observe the individual erythrocyte.

In this paper the features of the apparatus will be described in detail and representative data on human blood will be shown for the purpose of comparison with the results obtained by other techniques. The kinetic model and equation of the erythrocyte aggregation will be presented in the accompanying paper (28). The clinical application on the diabetes mellitus will be published separately.

MATERIALS AND METHODS

Materials

Fresh heparinized blood was drawn by venipuncture from healthy donors; the amount of heparin was 40 U/10 ml blood. Plasma was separated by centrifugation (3,000 rpm for 5 min at 5°C) and further centrifuged (15,000 rpm for 20 min at 5°C) to remove platelets. The erythrocytes were washed twice with the isotonic phosphate-buffered saline solution (14) containing 0.25% human serum albumin and 5 mM D-glucose (pH 7.4) and then resuspended in the same solution to give hematocrit of ~10%.

Serum was collected after the coagulation of blood (without heparin). Human fibrinogen (90% clottable) was obtained from Kabi. Human γ-globulin, α-globulins, and albumin (defatted, crystallized) were purchased from Miles Laboratories. Heparin was a product of Novo Industri.

The fractionation of plasma proteins was carried out

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with the cellulose acetate membrane electrophoresis [in pH 8.8 barbital sodium buffer, ionic strength (I) 0.075, for 30 min at 5°C], and the percentages of proteins were measured by an automatic densitometer (Jookoo Sangyo model Densitron-20M, Tokyo).

Apparatus

As shown in Fig. 1, the apparatus consisted of *1)* an inverted microscope (Olympus Optics type IMT, Tokyo) with × 20 objective lens, halogen lamp, and 577-nm interference filter, *2)* a cone-plate viscometer (Tokyo Keiki type B, Tokyo) with transparent 0.8° cone (polyacrylate), *3)* a particle analyzer (Toyo Ink model Luzex-450, Tokyo) with TV camera (Karnicon type), and *4)* a desk-top computer (Hewlett-Packard model HP-85, Palo Alto, CA). Additionally, a motion analyzer (Sony model SVM-1110, Tokyo) was connected as a monitor. The counts (numbers) and total areas of the particles in the TV images were encoded and transferred to the computer at maximum 4 TV images/s. The rate of the cone rotation can be varied in eight steps. For routine operation the shear rate of 7.5 s^{-1} was adopted (see RESULTS).

Manipulation

The washed erythrocytes were mixed with autologous plasma (or diluted plasma with isotonic phosphate-buffered saline) in a test tube (final hematocrit <0.6%); the sample (0.6 ml) was poured into the gap between the cone and glass plate; then (if needed) the focus of microscope and the grey level of the analyzer were readjusted. The microscopic focus was set to a few microns above the top surface of the plate, thus *1)* the sedimented erythrocytes attached to the bottom (if any) were not measured, and *2)* the upper portion (close to the moving cone) was invisible: e.g., in routine operation the position of the observation was fixed at 10 mm away from the cone center, where the gap depth was ~140 μm and the moving erythrocytes in the bottom portion (range ~30 μm) could be observed. The room temperature was kept at 25 ± 1°C. The pH of the plasma was measured after the experiment using a pH meter with a capillary electrode (Radiometer model PHM 64, Copenhagen, Denmark) at 25°C.

Software

During operation the sets of data (counts and total areas of particles) were consecutively encoded and transferred to the computer and monitored on a HP-85 screen as a graphic display of area/count vs. time, with the sampling rate reduced to intervals of ~1.3 s. In routine operations 800–1,200 sets of data were processed in 16–26 min. The raw data were either stored on the cartridge tape or displayed as the graphs of count vs. time, area vs. time, area/count vs. time, etc. (see Fig. 3). For the presentation of the time courses, the sequential means and standard deviations in 25 or 30 s (i.e., 19 or 24 TV images) were routinely processed and displayed.

RESULTS

Time Course of Erythrocyte Aggregation

A representative time course of the aggregation process is shown in Fig. 2. At the first stage (Fig. 2*A*, left) the one-dimensional aggregates (rouleaux) were gradually forming. In the next stage (Fig. 2*B*, left) the rouleaux grew slowly. Finally, the large three-dimensional aggregates developed from rouleaux (Fig. 2*C*, left), but after the passage of large aggregates only a few erythrocytes were seen (Fig. 2*D*, left). During these processes the erythrocytes gradually sedimented from the upper portion (invisible, out of focus) to the bottom portion (in the range of microscopic focus). The images processed by the particle analyzer are shown in the right side of Fig. 2; the count and area of the white spots are encoded.

The computed time courses are shown in Fig. 3: the time courses of counts, total areas, and area/count, expressed by the sequential means ± SD (of 19–20 TV images, i.e., in the period of 25 s), are presented.

The aggregation process can be distinguished in three stages as follows. *1)* In the first stage (marked as I, Fig. 3), both the count and area increased and the area/count slightly increased, i.e., the erythrocyte sedimentation and the development of the one-dimensional short rouleaux

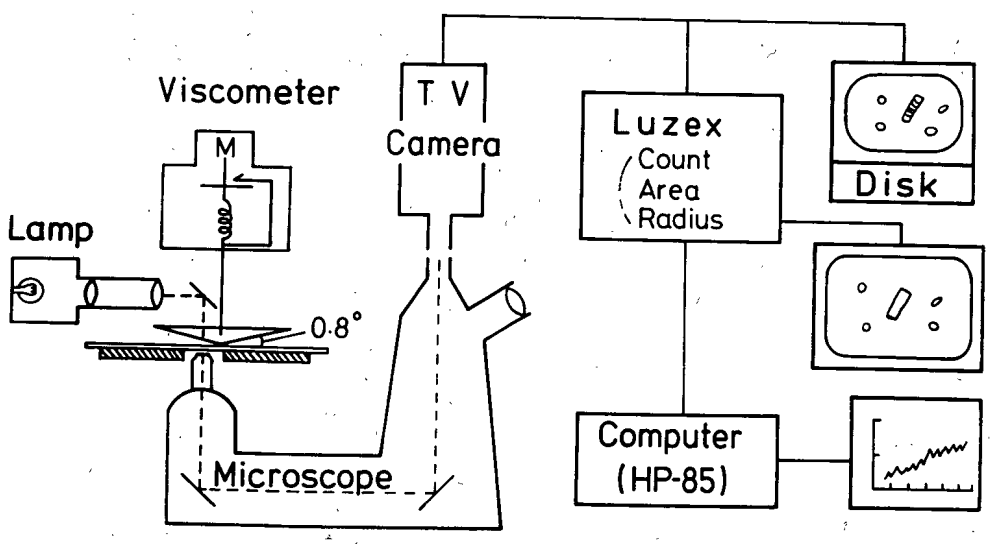

FIG. 1. Diagram of apparatus (see METHODS AND MATERIALS.)

(composed of less than several erythrocytes) were dominant, which corresponds to Fig. 2*A*. *2*) In the second stage (marked as II, Fig. 3), the count decreased but the area scarcely altered; thus the area/count gradually increased with slight increase of the standard deviation, i.e., the one-dimensional long rouleaux (of more than several erythrocytes) grew, which corresponds to Fig. 2*B*. *3*) In the third stage (marked as III, Fig. 3), both the count and area decreased in nearly exponential fashion and the area/count increased with augmenting standard deviation, i.e., the three-dimensional aggregates developed and the separation between the aggregates and plasma occurred, which corresponds to Fig. 2, *C* and *D*.

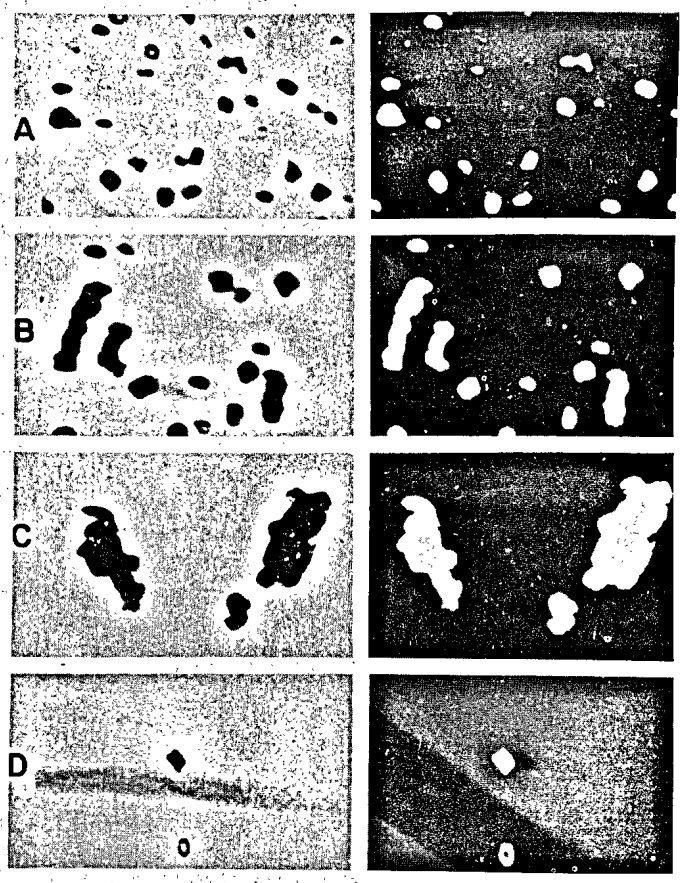

FIG. 2. Representative time course of aggregation process of erythrocytes. *Left side*: TV images before processing by the particle analyzer. *Right side*: TV images after processing by the particle analyzer. *A*: first stage; *B*: second stage [formation of one-dimensional aggregates (rouleaux)]; *C*: third stage (formation of large and three-dimensional aggregates); and *D*: third stage (after passage of large aggregates). Experimental conditions were 100% autologous plasma, 25°C, hematocrit 0.26%, depth 140 μm, and $\dot{\gamma}$ 7.5 s^{-1}.

The increment of the computed area/count in the second stage is adopted as a measure of the rate of rouleaux formation, expressed in terms of μm^2/min. However, the "area" itself does not exactly correspond to the number of erythrocytes because of tumbling motion (10). The following situation should be noted, taking the radius (r) and the mean thickness (d) of erythrocytes. *1*) The erythrocytes are rotating in shear flow, so that their area varies considerably with the angle, e.g., for a parallel position the area (πr^2) is $\sim$45 μm^2 and for a perpendicular position the area ($2rd$) is $\sim$12 μm^2. If the computed values of area/count are extrapolated to *time 0*, the estimated area of a single erythrocyte of 25–30 μm^2 is obtained, which seems to agree with the angular average of area of a single erythrocyte. 2) The long rouleaux are usually rotating along their longitudinal axis, thus the area of the individual erythrocytes in rouleaux may be close to the minimal value ($2rd$). *3*) Some of the short rouleaux (of n erythrocytes) are turning freely, thus the area of rouleaux varied from $2nrd$ to πr^2 depending on the angles. *4*) When the three-dimensional aggregates appear, the observed area no longer corresponds to the number of erythrocytes in the aggregates because of overlap. Overall, the observed area is related to the number of erythrocytes, and the computed area/count gives a rough measure of the averaged area of a particle unless three-dimensional aggregates are developed.

Effect of Various Factors on Rate of Rouleaux Formation

Taking the increment of the area/count as a measure of the rate of rouleaux formation, the factors affecting the aggregation process were briefly surveyed to find the appropriate experimental conditions and validate the present method (Fig. 4).

Shear rates. When the shear rate was more than 38 s^{-1}, the encoding of count and area became difficult because the erythrocytes moved too rapidly. On the other hand, if the shear rate was less than 1.9 s^{-1}, the erythrocytes moved so slowly that the sedimentation became dominant (and the frequency of collision decreased) with the distinction between the aggregates and the overlap of the erythrocytes becoming difficult. In the shear rate

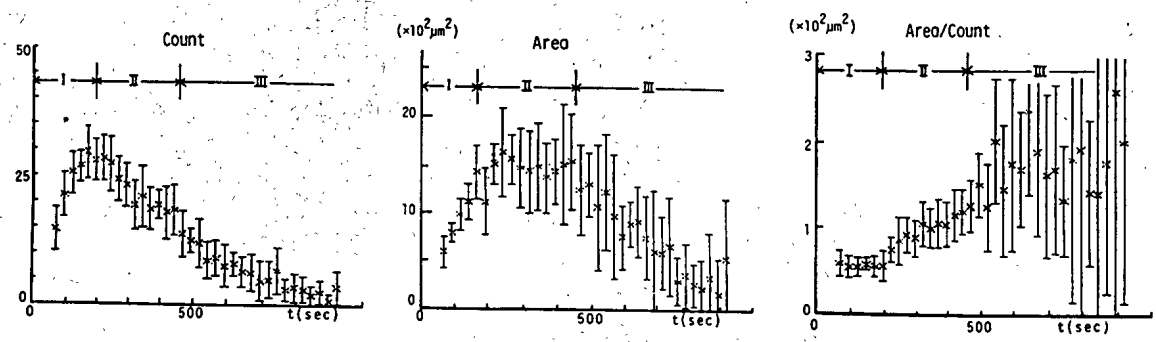

FIG. 3. Time courses of the counts, area, and area/count during erythrocyte aggregation. Stages of aggregation process (I, II, and III) correspond to 3 stages (see RESULTS). Experimental conditions were same as in Fig. 2.

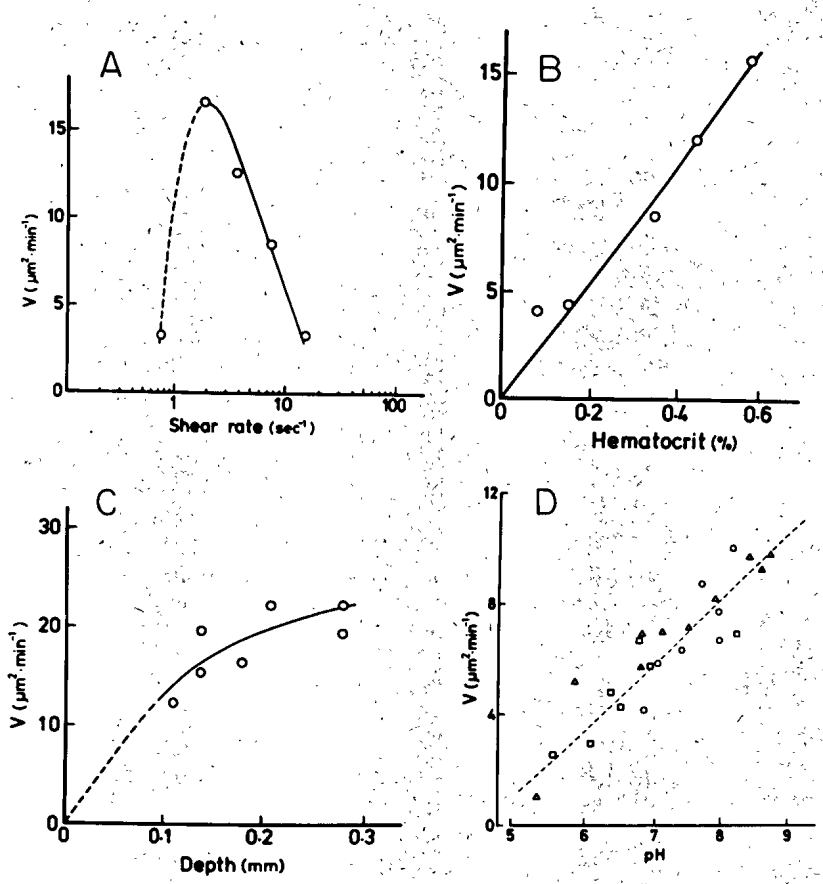

FIG. 4. Effect of various factors on rate of rouleaux formation. Conditions are given for each. *A*: effect of shear rate with diluted autologous plasma 70%, 25°C, depth 140 μm, hematocrit 0.26%, and pH ≃ 7.9. *B*: effect of hematocrit with diluted autologous plasma 70%, 25°C, depth 140 μm, $\dot{\gamma}$ 7.5 s^{-1}, and pH ≃ 7.9. *C*: effect of depth with diluted autologous plasma 70%, 25°C, $\dot{\gamma}$ 7.5 s^{-1}, hematocrit 0.34%, and pH ≃ 7.9. *D*: effect of pH with diluted autologous plasma 70%, 25°C, depth 140 μm, hematocrit 0.26%, and $\dot{\gamma}$ 7.5 s^{-1}. The symbols ○, △, and □ represent different series of expts.

range ($1.9 \leq \dot{\gamma} \leq 15\ s^{-1}$) the rate of rouleaux formation increased with decreasing shear rates (Fig. 4*A*). A shear rate of 7.5 s^{-1} was employed routinely for the studies.

Hematocrit. To avoid the overlap of the erythrocyte images, the hematocrit should be less than 0.6% at the location 10 mm away from the cone center (depth of ca. 140 μm). Under the routine conditions ($\dot{\gamma} = 7.5\ s^{-1}$ in 70% diluted plasma) the rate of rouleaux formation was proportional to hematocrit up to 0.6% (Fig. 4*B*). A hematocrit of 0.25% was used in the routine studies.

Depth. By changing the distance between the cone center and the position of microscopic observation, the gap width could be varied (~110–280 μm). The results are shown in Fig. 4*C*. At increasing gap width more erythrocytes sedimented so that the rate of rouleaux formation, as well as that of three-dimensional aggregation, increased. In routine operations a depth of 140 μm was used.

Anticoagulant (heparin). The concentration of heparin was increased from 40 to 500 U/10 ml, but the rate of rouleaux formation in 70% plasma was unaffected, confirming an earlier observation (25).

pH. To vary the pH, the plasma was diluted to 70% with the isotonic mixture of KH_2PO_4 and K_2HPO_4 (or H_3PO_4). The rate of rouleaux formation increased at higher pH, as shown in Fig. 4*D*. When the pH was lowered below 6.3, the erythrocytes tended to lose their biconcavity (due to swelling), which may have led to the lower rate of rouleaux formation.

Plasma proteins. It is well known (2–4, 6–8, 13, 16, 17, 19, 24, 27, 31, 32) that plasma proteins affect the rouleaux formation. In serum and isotonic NaCl solution the aggregation was scarcely developed, as already reported (23). However, an addition of either fibrinogen, γ-globulin, or α-globulins to 70% plasma facilitated aggregation.

1) In 60% diluted serum the aggregation hardly occurred (Fig. 5*A*), as well known (22). The count and area increased at first (because of the sedimentation), then decreased gradually (because of the sedimentation and adhesion to the bottom plate), but the area/count remained unchanged throughout. *2)* In diluted plasma three stages could be recognized as described above (Figs. 3 and 5*B*). *3)* When γ-globulin was added to 70% plasma the three-dimensional aggregates were formed very quickly (Fig. 5*C*). The long one-dimensional rouleaux (of more than several erythrocytes) was not formed, but three-dimensional aggregates developed promptly from the short rouleaux (of 3–5 erythrocytes); thus the decrease of the count and area was rapid and SD of the area/count became very large. *4)* Effect of fibrinogen added to 70% diluted plasma was shown in Fig. 6. The rate of rouleaux formation increased proportionally to the fibrinogen concentration up to 600 mg/dl. Fibrinogen seemed to be less effective for the formation of three-dimensional aggregates than γ-globulin. The distinction between the effect of fibrinogen and γ-globulin on rat blood is examined in the accompanying paper (28).

Shape of erythrocytes. When the washed erythrocytes were kept at room temperature for several hours, they changed shape to echinocytes or stomatocytes (depending on blood donors). The rate of rouleaux formation decreased as the content of these transformed cells increased. Therefore, it is most important for the quanti-

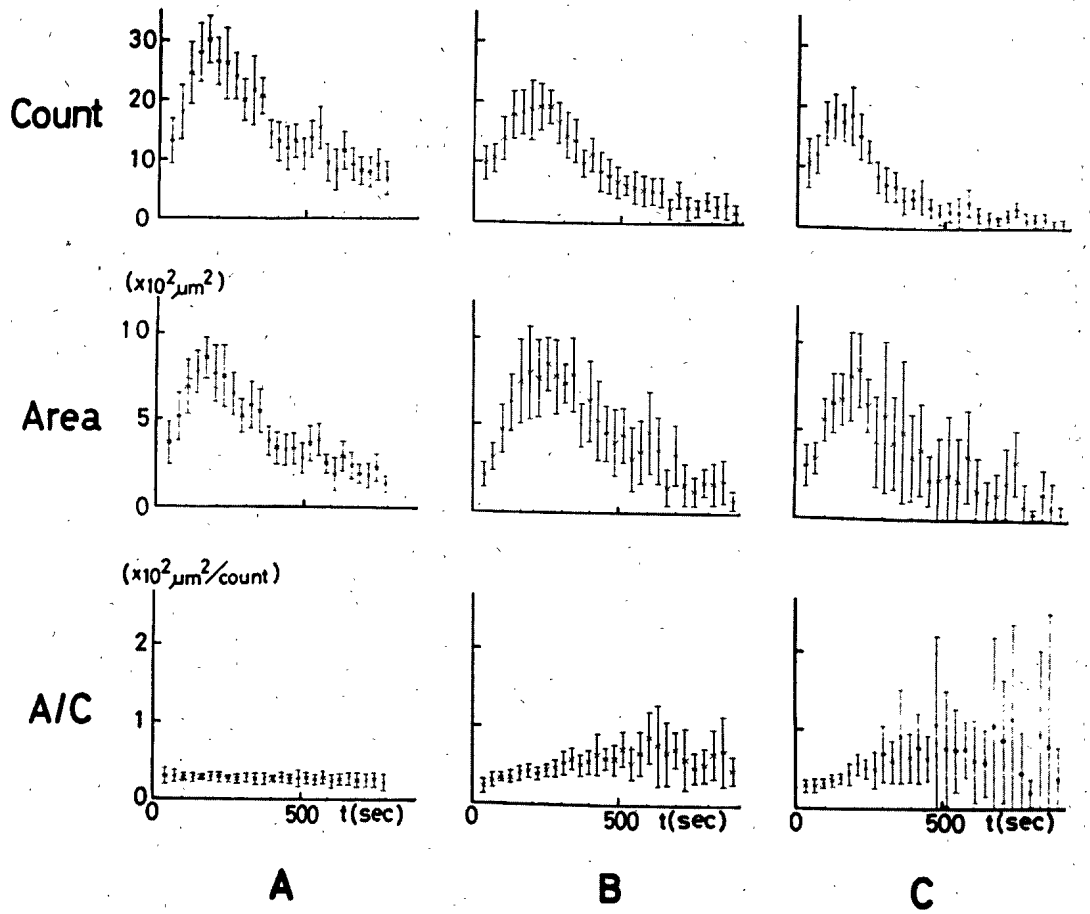

FIG. 5. Time course of counts, area, and area/count (A/C) during erythrocyte aggregation in different media. *A*: 60% diluted autologous serum; *B*: 70% diluted autologous plasma; and *C*: 70% diluted autologous plasma + γ-globulin (final concn 0.6 g/dl). Conditions were 25°C, hematocrit 0.26%, $\dot{\gamma}$ 7.5 s^{-1}, depth 140 μm, and pH ≃ 7.9.

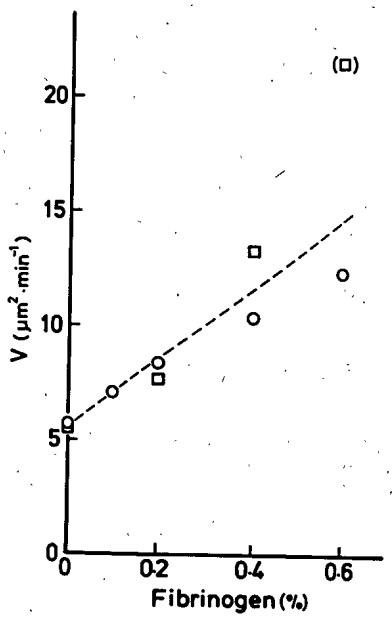

FIG. 6. Effect of various fibrinogen concentrations on rate of rouleaux formation of erythrocytes. Symbols are ○, blood of *KI*; □, blood of *KK*. Conditions were 70% diluted plasma, 25°C, $\dot{\gamma}$ 7.5 s^{-1}, hematocrit 0.26%, depth 140 μm, and pH ≃ 7.9.

tative measurement of the aggregation process to use fresh erythrocyte samples before any shape transformation has occurred.

Variation of Rate of Rouleaux Formation

Because the rate of rouleaux formation is highly dependent on plasma constituents and on erythrocyte properties, the rate varied from sample to sample.

Daily variation. The data of one donor (*KI*) during 0.5 yr were gathered. The variation of the rates of rouleaux formation in 70% diluted autologous plasma was ±13%.

Individual variation. The variation of the rates in 70% diluted autologous plasma of 11 healthy donors was ~17%; the composition of plasma proteins quantified with the cellulose-acetate electrophoresis differed from person to person (Table 1).

DISCUSSION

Characteristics of the Method

The reversible aggregation of the erythrocytes in plasma is related to *1*) the blood viscosity at low shear rates, *2*) the sludging or clumping of erythrocytes in microcirculation, and *3*) the erythrocyte sedimentation rate. Therefore, a kinetic study of the aggregation process is important for an understanding of these phenomena. The present method can observe the individual erythrocyte moving in the microscopic field under controlled shear conditions, thus it is adequate for the kinetic study. The overall process can be described in three stages, especially in diluted plasma: *1*) erythrocyte sedimenta-

TABLE 1. *Variation of rouleaux formation rate of human erythrocytes, hematological indexes, and plasma proteins*

Variable	Values
Rouleaux formation rate*, μm^2/min	5.2 ± 0.9
Erythrocyte	
MCV, μm^3	90.3 ± 3.1
MCH, pg	30.6 ± 1.1
MCHC, g/dl	34.0 ± 0.5
Plasma	
Total protein, g/dl	7.2 ± 0.6
Protein composition, %	
Albumin fraction	58.6 ± 3.8
α_1-Fraction	4.1 ± 1.0
α_2-Fraction	6.6 ± 1.4
β-Fraction	10.1 ± 1.1
γ-Fraction	10.7 ± 2.2
ϕ-Fraction	9.7 ± 2.9

Values are means ± SD; n = 11. MCV, MCH, and MCHC, mean corpuscular volume, hemoglobin, and hemoglobin concn, respectively. * Hematocrit 0.26%, in 70% diluted autologous plasma, $\dot{\gamma}$ 7.5 s^{-1}, at 25°C.

tion, *2*) one-dimensional rouleaux formation, and *3*) formation of three-dimensional aggregates. As shown in Fig. 3, the rate of rouleaux formation can be easily estimated taking the increment of the computed area/count, unless the sequential standard deviation becomes too large. When only one-dimensional aggregates (rouleaux) are formed without the development of three-dimensional aggregates, computer simulation is successfully made on the basis of a kinetic model similar to actin polymerization [see companion paper (28)]. From a technical point of view the present method has certain advantages, e.g., the automated data processing and simplicity of data presentation.

On the other hand, the growth rate of three-dimensional aggregates is hardly defined from the encoded data, although the decay of the count and area apparently followed the first-order kinetics at the third stage. Also our apparatus cannot be applied to whole blood, i.e., the overlap of the erythrocytes interferes with the rate determination. The routine procedures were carried out at the lower hematocrit (<0.6%) in 70% diluted plasma: as the ratio of the erythrocyte numbers to plasma proteins is very low compared with the whole blood, the observed kinetic parameters are an indirect measure for whole blood, but they reflect and may amplify the basic properties.

There is individual variation in the rates, which is dependent on blood donors, because the composition of plasma proteins varies from individual to individual (Table 1), although the daily variation of the rate is reasonably small. In one case of multiple myeloma and 12 cases of diabetes mellitus with proliferative retinopathy, the average rate increased (unpublished data), suggesting that the present method will be applicable to clinical studies.

Various Factors Affecting Rouleaux Formation

The rate of rouleaux formation is influenced by various factors; some of them have already been discussed by many authors (2, 3, 6, 8, 16, 17, 19, 22, 24, 25, 30–32). We confirmed most of the previous results and obtained some new information.

Shear rate. The shear rate gave dual effects on the rate of rouleaux formation, as shown in Fig. 4*A*. If the shear rate was too low ($\dot{\gamma} < 1.9\ s^{-1}$) the erythrocytes sedimented and barely collided to form rouleaux, but if the shear rate was too high ($\dot{\gamma} > 38\ s^{-1}$) the aggregation hardly developed. Thus the rate of rouleaux formation was highest at the optimal shear rate ($\dot{\gamma} \simeq 1.9\ s^{-1}$ at hematocrit of 0.26% in 70% diluted plasma). Further, if the shear rate was suddenly increased after forming the aggregates, the aggregates were disrupted and the area/count was decreased.

The shear-dependent formation of rouleaux has been proposed to explain some phenomena, e.g., the biphasic behavior of light transmission observed with the rheoscope (13), the increased viscosity at low shear (1, 4), and the biphasic behavior of the erythrocyte sedimentation rate in Cuette flow (5, 20). Most of these phenomena were observed at hematocrit of 30–50%. Thus the optimal shear rate for these phenomena differed from our results obtained at low hematocrit.

In addition, the centrifugal movement of the rotating particles may occur to some extent. However, the centrifugal force ($m \cdot \omega^2 \cdot r$) on erythrocyte, even at the position close to the rotating cone (mass, m; radius, r; angular velocity, ω), is far less than the gravity force ($m \cdot g$) and thus has no significant effect on the measurement.

Hematocrit. The dependence on hematocrit is certainly due to the increased frequency of erythrocyte collision at higher hematocrit. The relation between the hematocrit and the rate of rouleaux formation is a complex and nonlinear function [see companion paper (28)], but a linear relation was observed (Fig. 4*B*), perhaps due to the low hematocrit.

The increase of the gap width increased the rate of rouleaux formation (Fig. 4*C*), because the hematocrit of the lower portion under study is augmented gradually by the erythrocytes undergoing sedimentation [see companion paper (28)].

Plasma proteins. The influence of plasma proteins on the erythrocyte sedimentation rate and on the low shear viscosity of blood have been extensively studied (2, 3, 6–8, 16, 17, 19, 22, 24, 25, 30–32). Further the clinical features of these proteins as the risk factors of the circulatory disturbances have been discussed (15, 29).

Among the plasma proteins, fibrinogen and γ-globulin showed an accelerating effect on aggregation. These high molecular weight proteins were bound to the erythrocyte surfaces and produced aggregation by intercellular bridging [i.e., surface bridging force (3)]. The present method not only confirmed the effect of these plasma proteins but also provided quantitative data (Figs. 5 and 6).

Erythrocytes. The shape of the erythrocytes affected the aggregation process. When echinocytes were increased by storage at room temperature in the isotonic solution, the rate of rouleaux formation decreased. The steps of rouleaux formation seem to occur in the following order: *1*) contact by collision, *2*) increase in the contact area by sliding rather than flipping (28), and *3*) face-to-face adhesion. The transformed erythrocytes

hardly attached to each other, perhaps because the contact area was too small and the geometrical fitting of surface thus became difficult.

pH. The effect of pH was striking. As the pH of plasma was increased, the rate of rouleaux formation increased. According to Chien and Jan (3) the major factors of rouleaux formation are the surface bridging force due to inserted proteins as the attractive force and the electrostatic repulsive force, shear force, and membrane bending force as the dissociative forces. When pH increases, *1*) the net negative charge due to sialic acid (p*K* 2.6) on the erythrocyte surface may scarcely change, whereas *2*) the binding of fibrinogen to the surface has been shown to decrease (pH < 7) or to remain constant (pH > 7) (21). Therefore, the increased rate of rouleaux formation at higher pH could not be explained by the subtle changes in the surface charge or in the protein binding. Actually, an earlier report (9) showed the biphasic behavior of the erythrocyte sedimentation rate; the rate was highest at pH 8.

Another possibility to explain the pH dependence of the rate of rouleaux formation is the shape changes of erythrocytes. The mean corpuscular volume increases on lowering the pH of the medium (11, 18) (which we confirmed). At pH values lower than 6.3 the swelling of erythrocytes became evident microscopically. However, in our studies the washed erythrocytes (kept at pH 7.4) were mixed with the diluted plasma of a given pH and placed in the apparatus (within 40 s); therefore, the equilibrium between intra- and extracellular pH would not be established and the volume would be changing during the measurement. At the moment we are not able to compare the velocity of aggregation and the velocity of volume change. However, the pH-dependent volume change, i.e., the shape change, may be plausible explanation for pH dependence of the rouleaux formation.

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